Stat Modeling of Wildfires Using Meterorological and Fire-Weather Indices

Math 444

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Full Linear Model

log(area+1) ~ X + Y + month + day + FFMC + DMC + DC + ISI + temp + RH + wind + rain

Initial Stepwise Model

(log(area+1) ~ X + month + DMC + DC + temp + wind

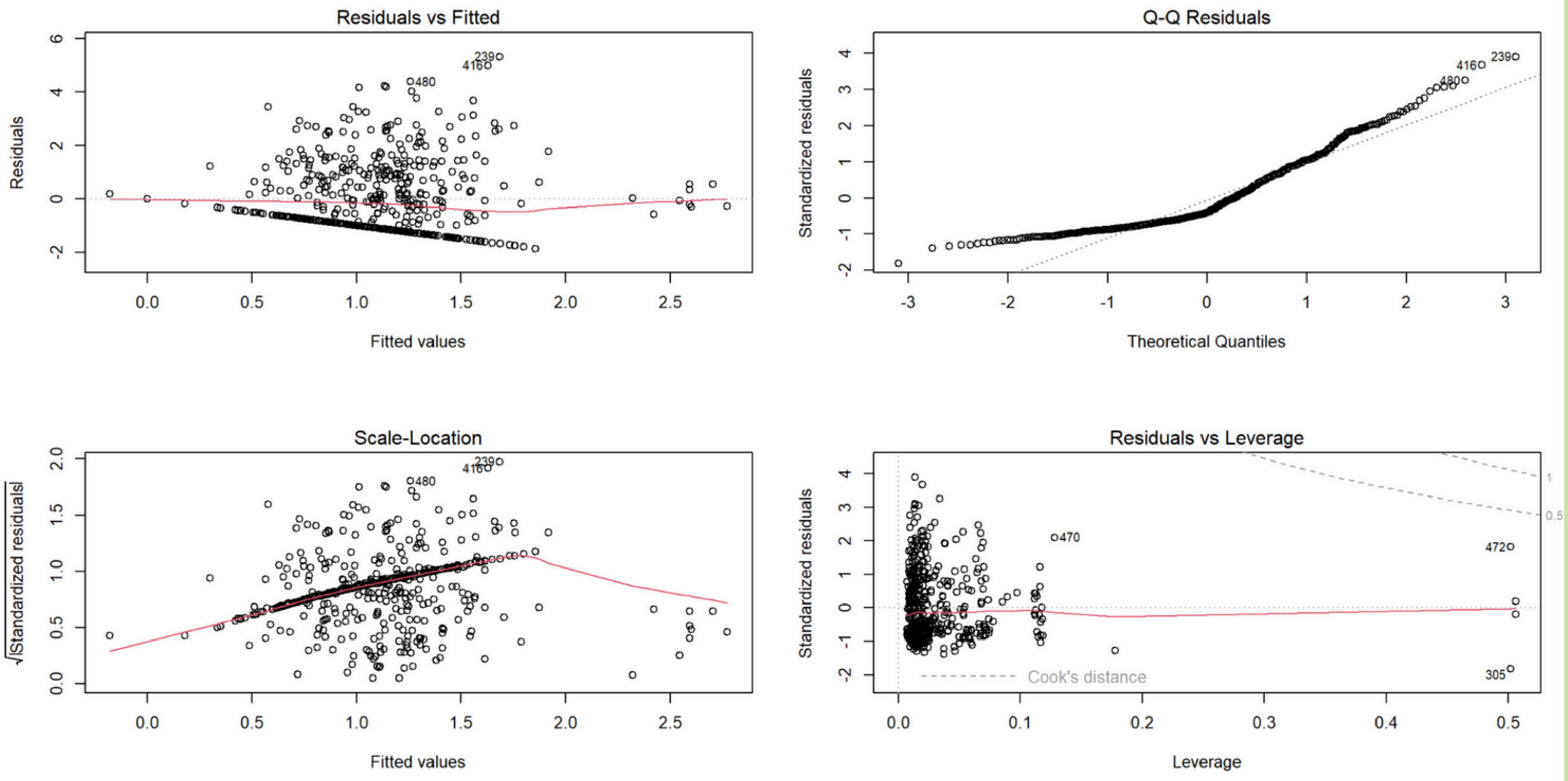
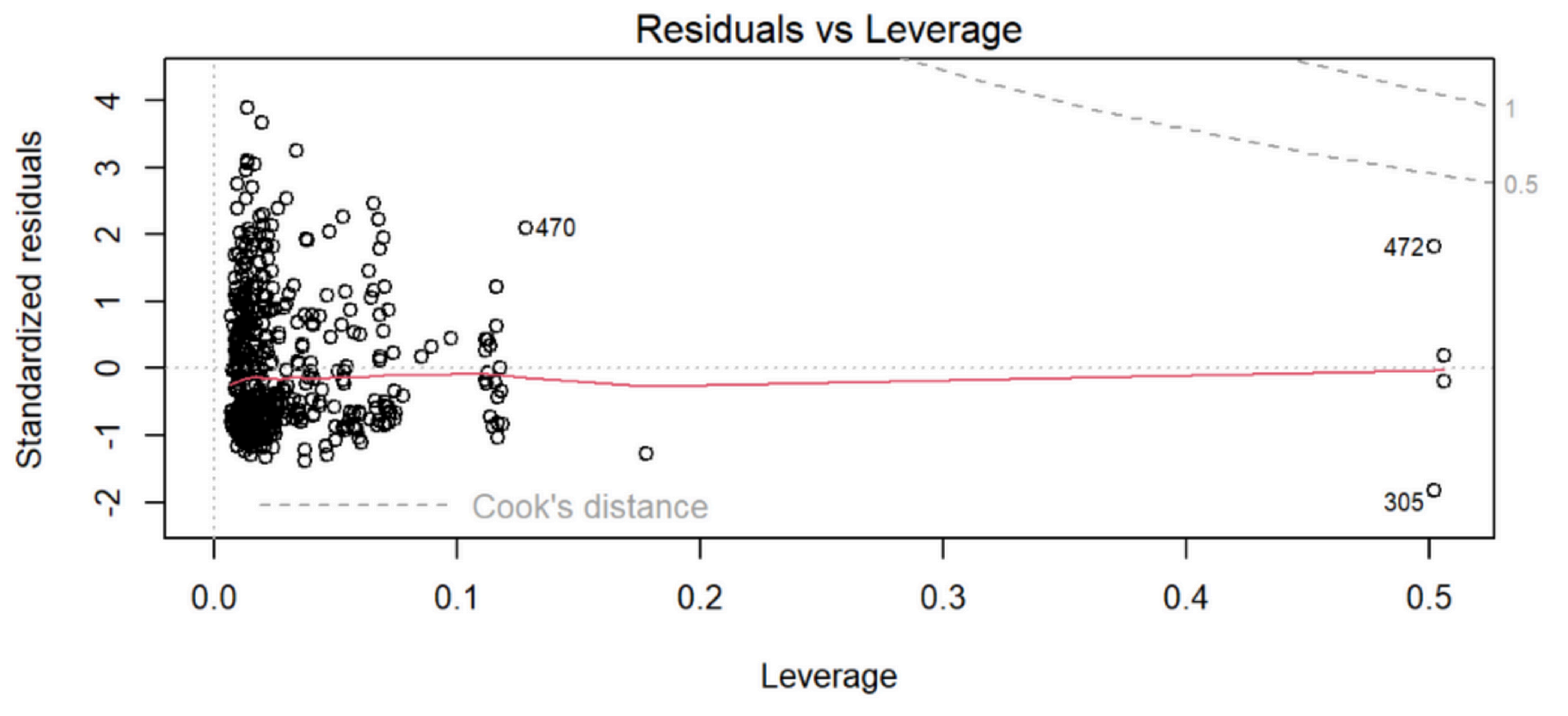


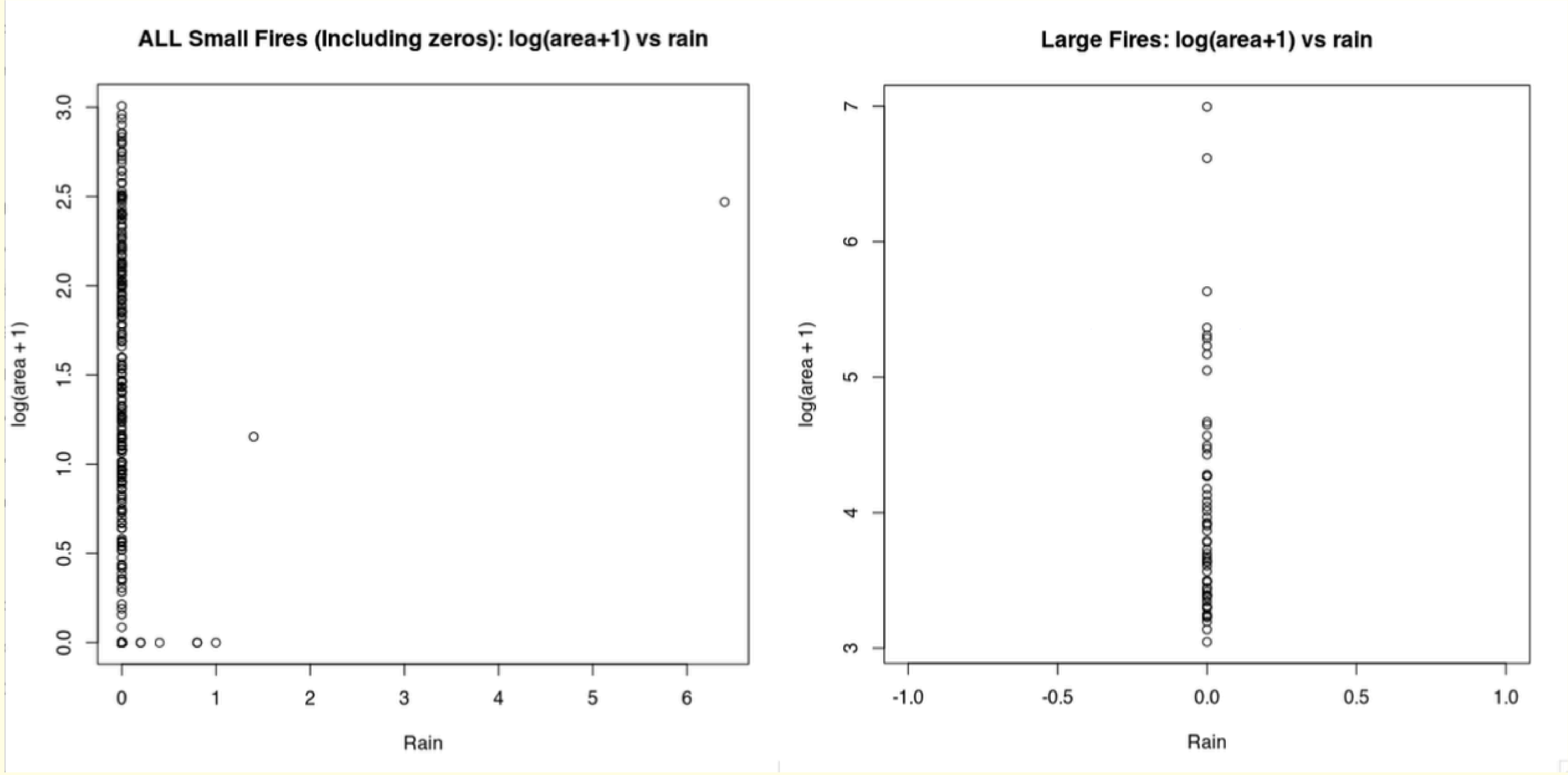
Figure 1: Diagnostic plots for full linear regression model

- R^2 adjusted is about 0.03 for the intitial, full model; predictors nearly explain none of the variation
- The Q-Q Residuals plot is right-skewed
- Scale Location shows heteroscedasticity

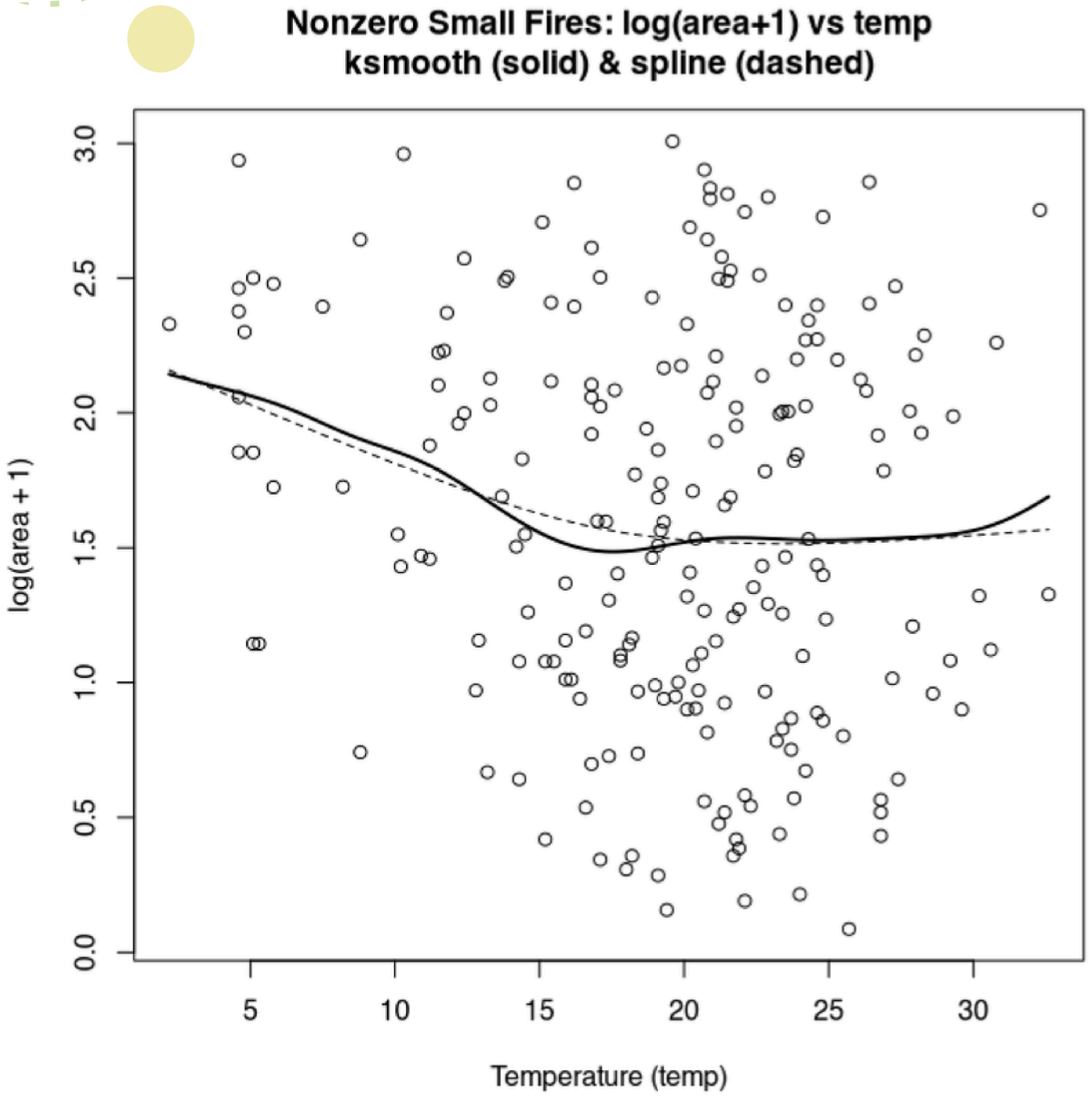
Small Fires vs Large Fires



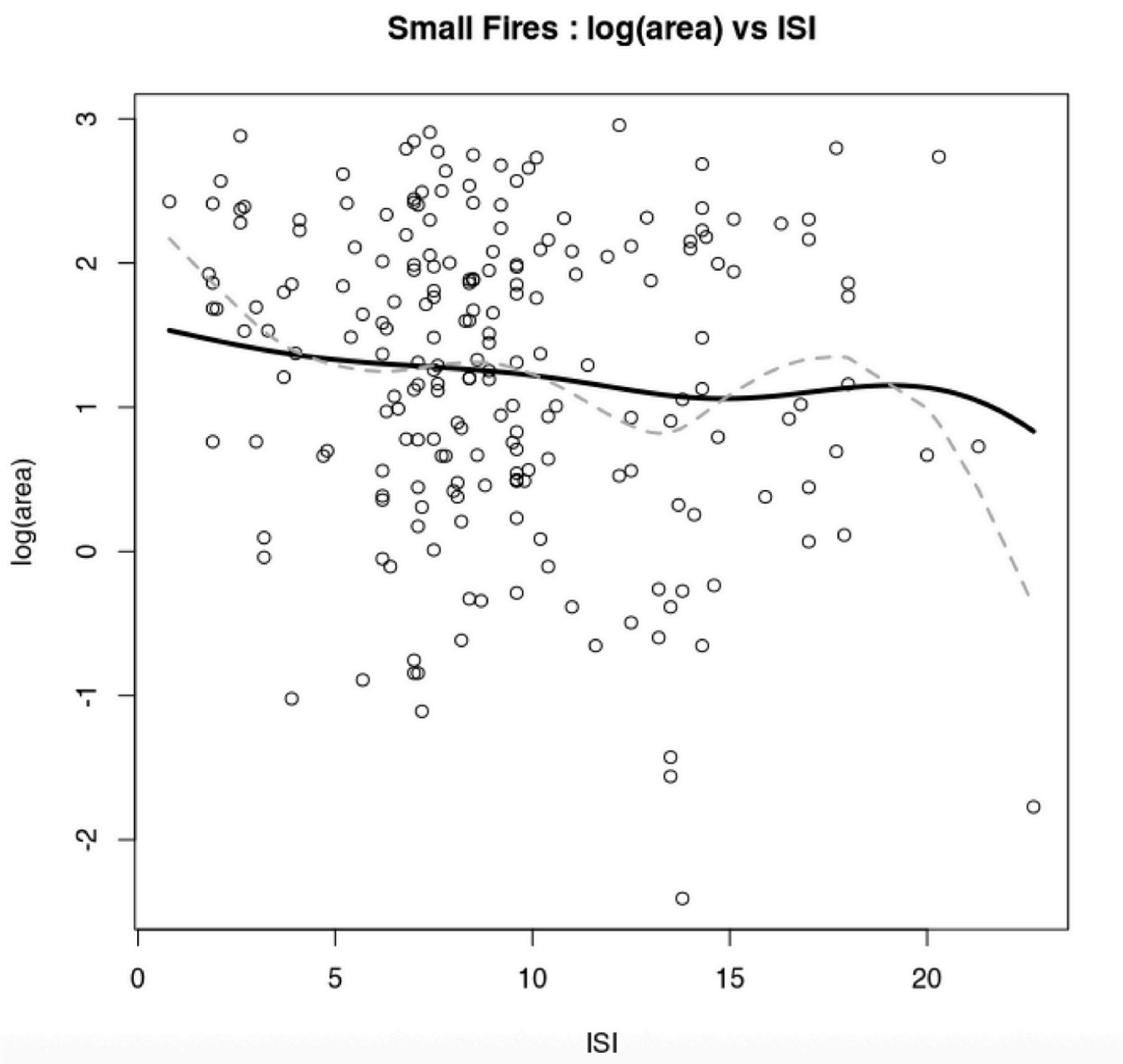
- In Residuals vs Leverage, there are two distinct clusters
- A separation between small and large fires around 20 hectares, matching Cortez and Morais claims
- We split the data into two, modeling each one separately



- Rain is almost always recorded as 0 at the start of a fire, matching common sense
- Rain has no variation so it doesn’t help explain the burned area

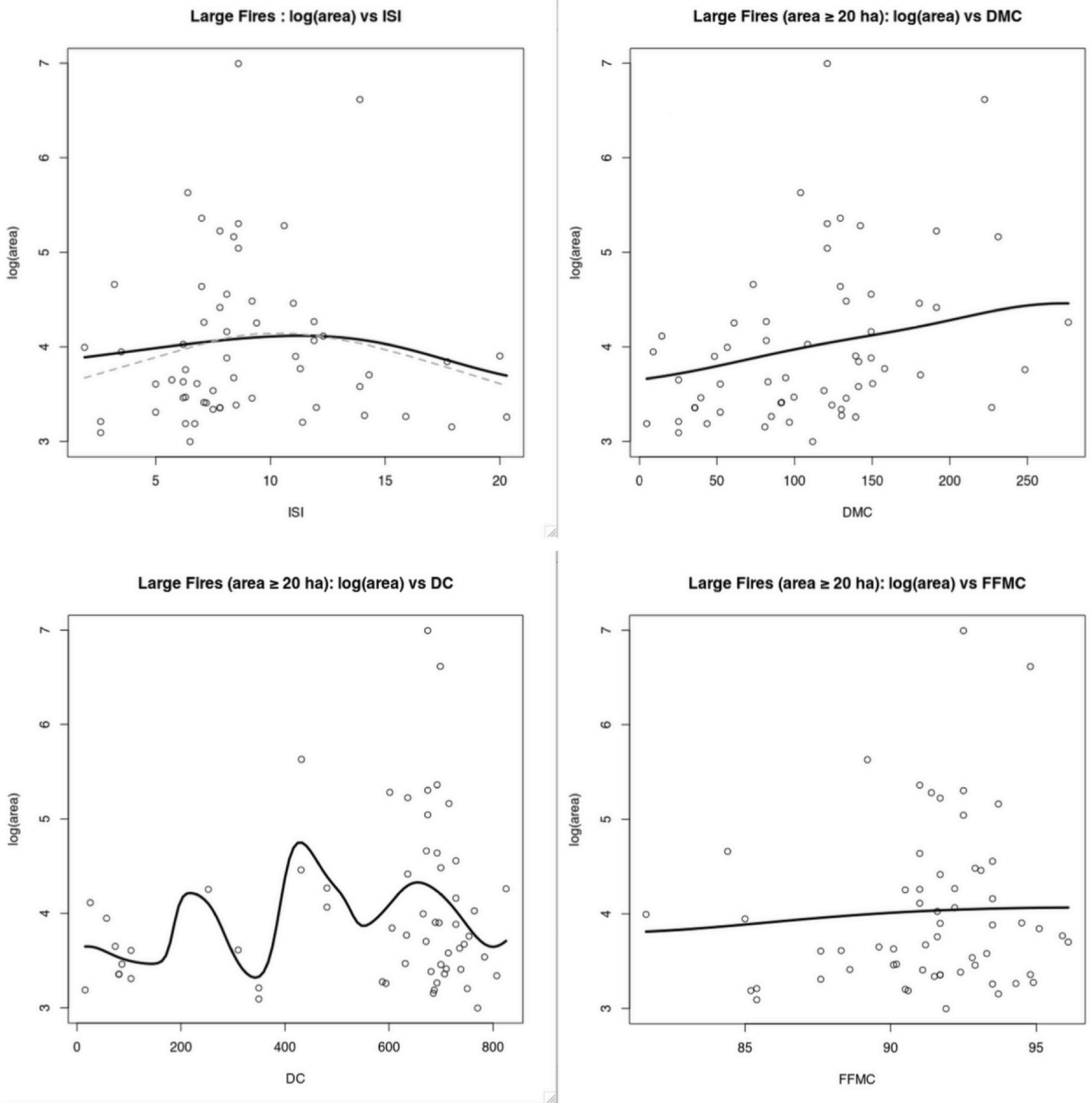


- This is one plot of the subset of small fires with temp, RH, wind, and rain versus log(area + 1)
- temp vs log(area + 1) displays a weak downward trend with large vertical variations at each temp
- After smoothing our plots, our R^2 adjusted is still around 1-3%, meaning weather variables don't explain small-fire size very well

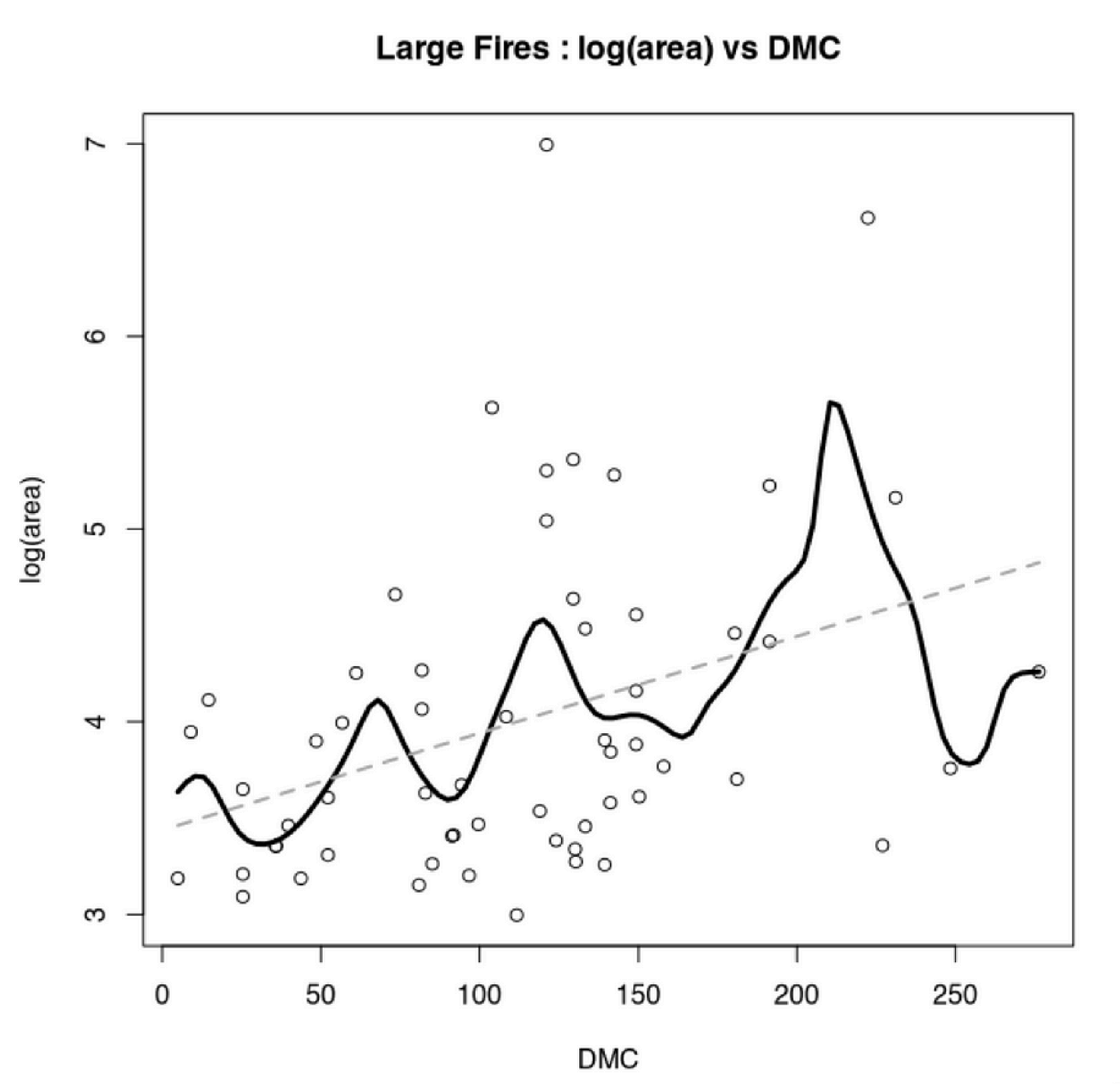


- When fitting a multiple regression model using FWI variables for small fires, our R^2 adjusted came out to 0.0218
- After completing a stepwise AIC procedure, our best model turned into a single-predictor regression focusing on ISI with its R^2 adjusted equalling 0.0229
- Using FWI variables for small fires is still not a good way to predict these events

Large Fires (FWI Metrics)

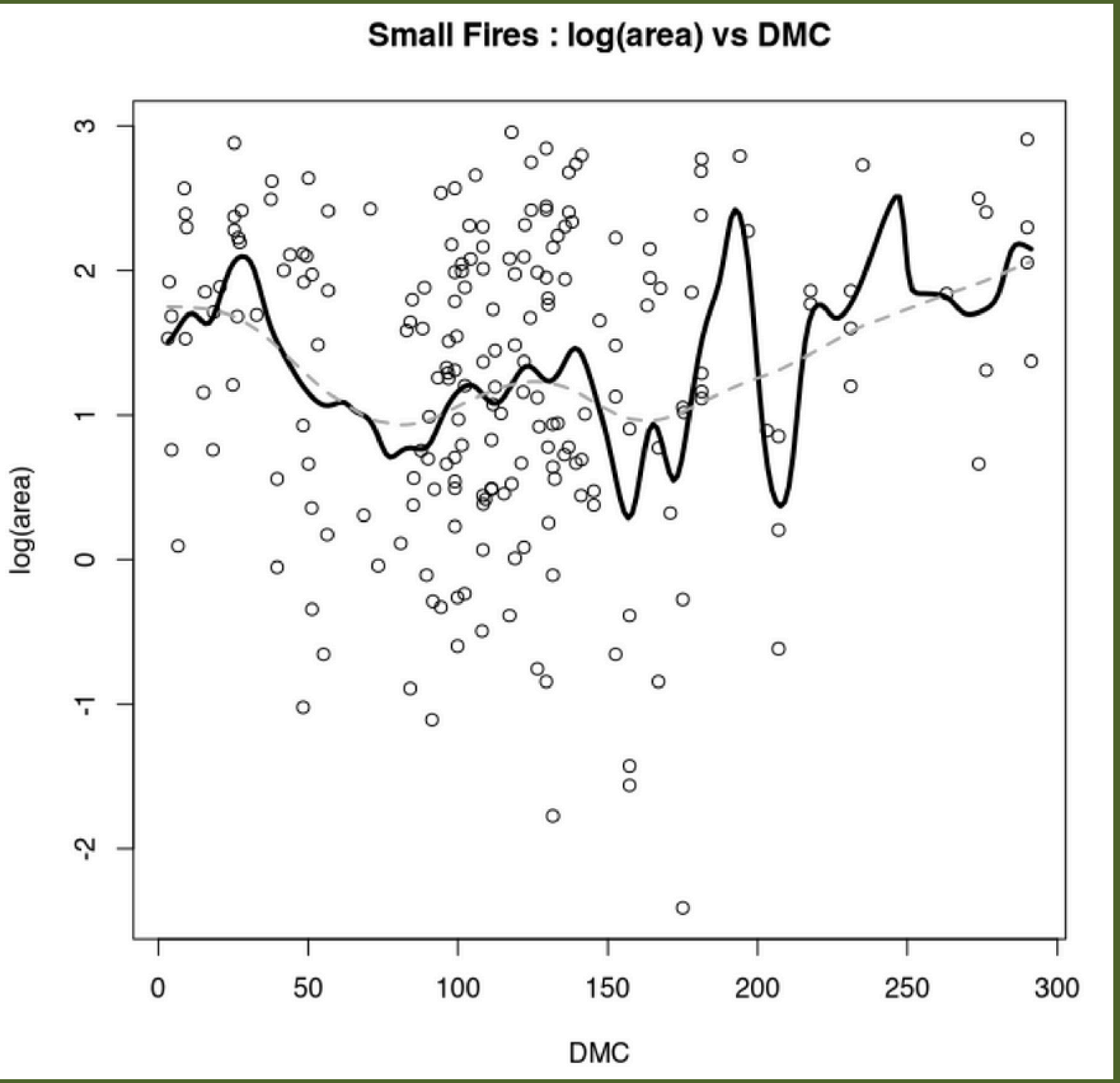


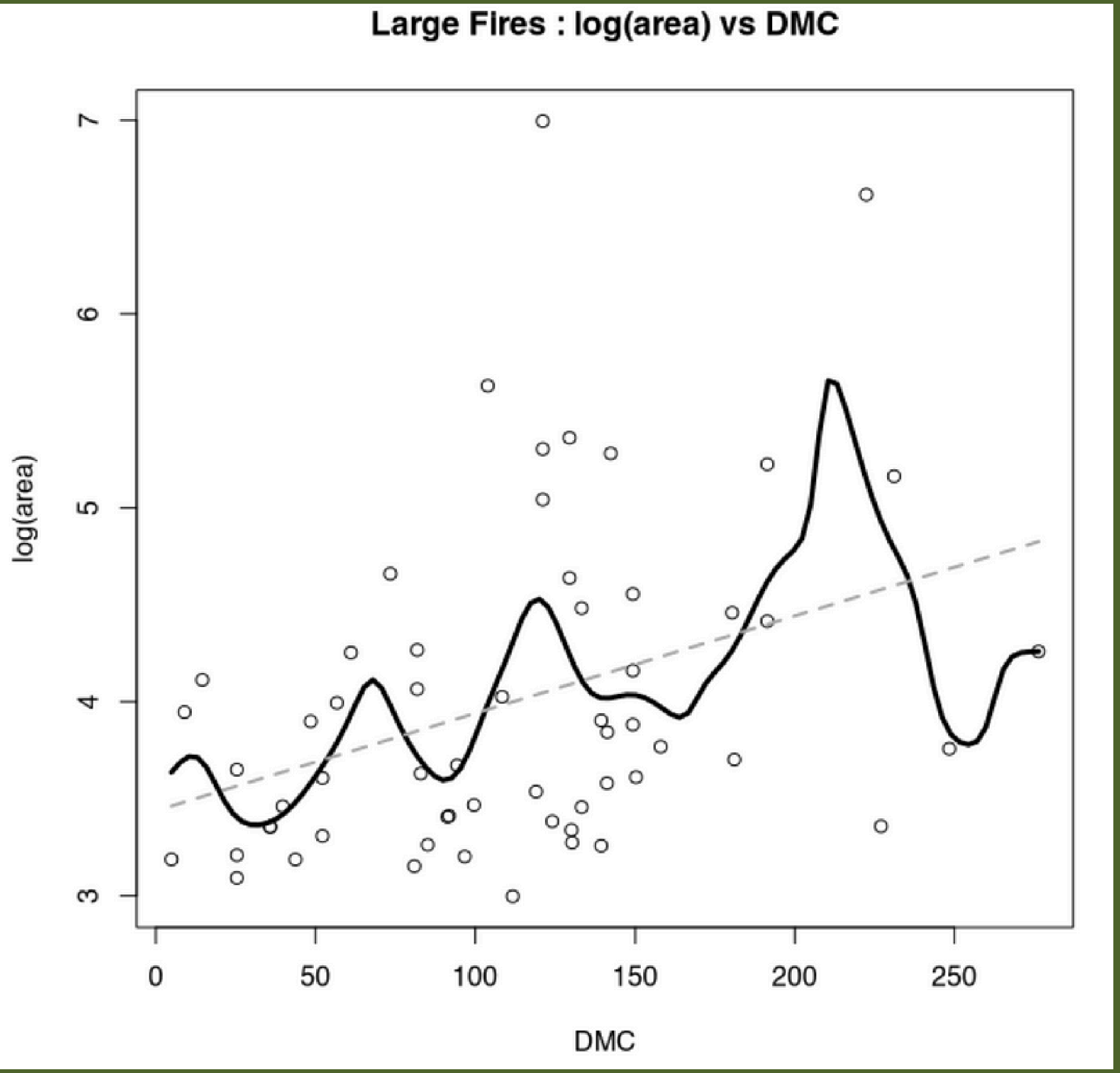
Large Fires (DMC)



- For large fires, DMC which measures medium-term fuel moisture, has an increasing noticable increasing relationship with log(area)
- As DMC increases, so does the area burned
- The R^2 adjusted gives us 0.12, which is much higher than anything for small fires

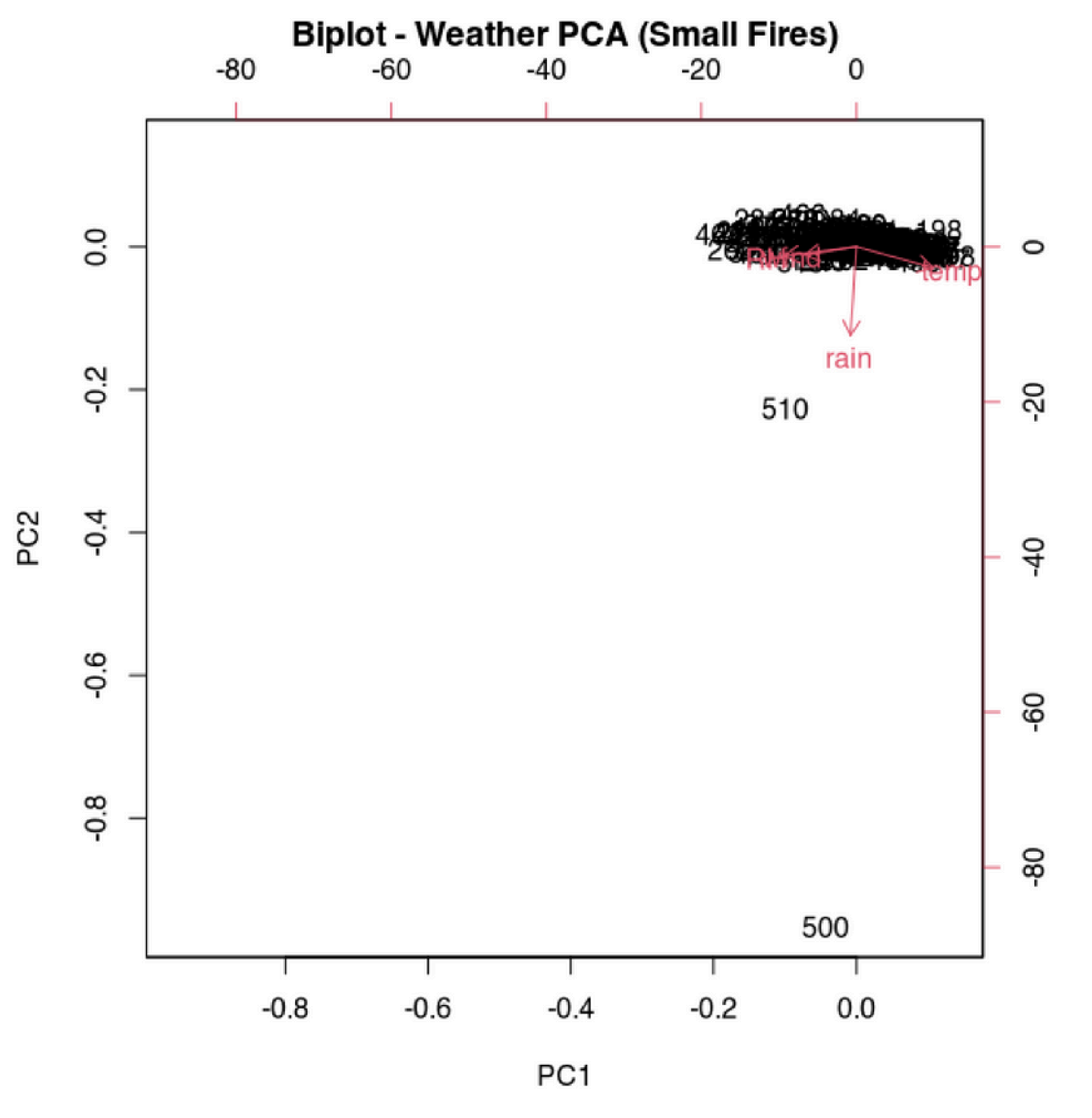
DMC: Small vs Large





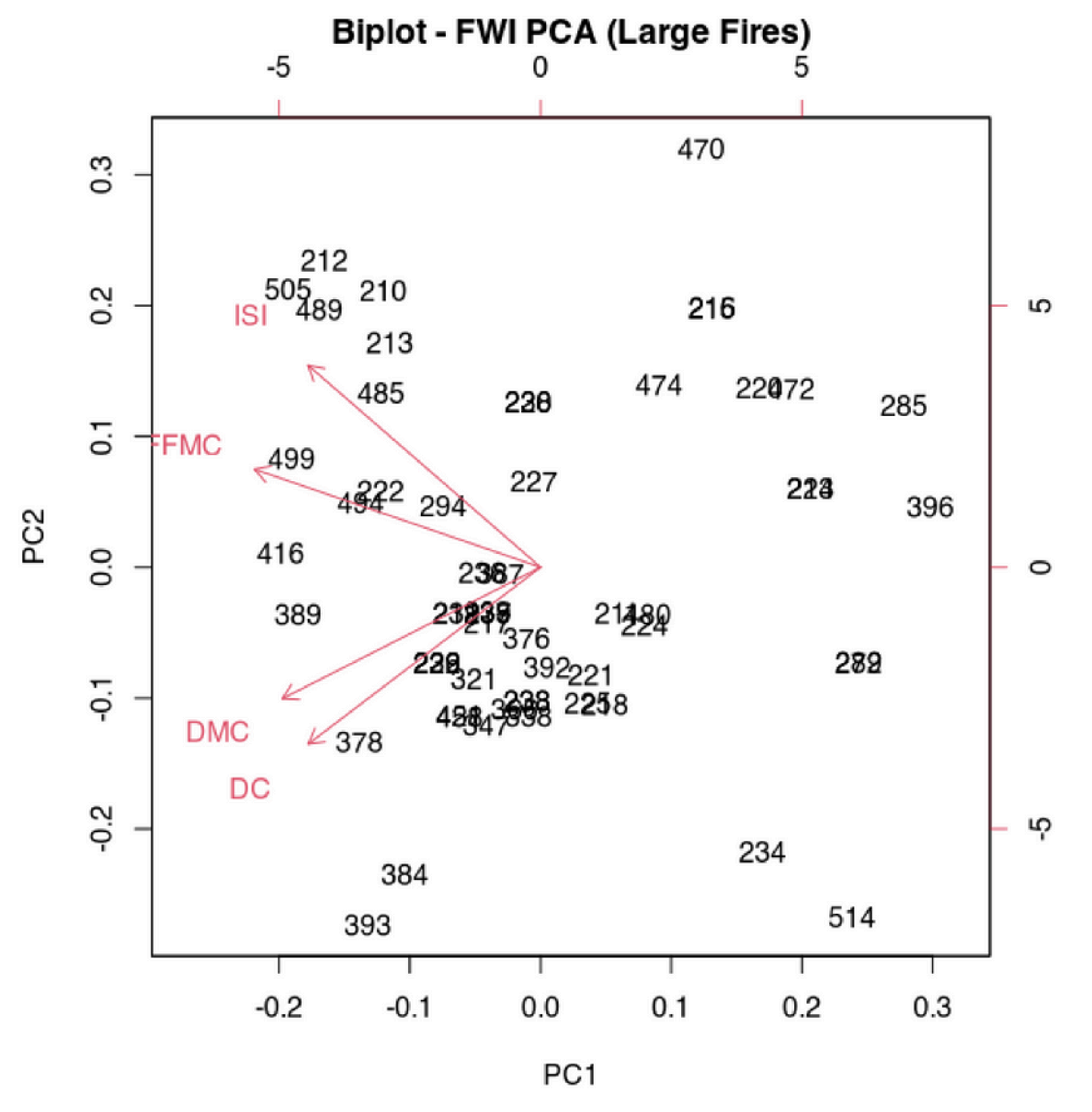
- DMC is a key predictor for large fires and very weakly related to small fires

PCA Small Fires



- PCA on weather variables gives first component 42% of variation explained, 68% total when added with PC2
- R^2 adjusted stays the same around 1-2%
- PCA reinforces earlier claims about small fires not having significant predictors

PCA Large Fires



- PCA shows all “dryness” FWI variables are highly correlated
- PC1 explains 62% of variance with 86% being the total with PC1 + PC2
- The R^2 adjusted comes out to 0.0649 which is consistent with earlier FWI regressions

Final Models

Final Small Fires Stepwise Model

(log(area+1) ~ X + DMC + DC + temp + wind

Final Large Fires Model

log(area) ~ FFMC + DMC + DC + ISI

CONCLUSION

- No variable was significantly enough to predict small fires
- Dry indices such as DMC and FWI PC1 were key to predicting large fires
- R^2 values are too modest, fitting the idea that wildfire behavior is complex and is influenced by many factors